Bevara Jagan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.53
Tirumala Educational Institute	Class XII (Andhra Pradesh Board)	2024	98.20%
Narayana E Techno School Vizianagaram	Class X (Andhra Pradesh Board)	2022	97.83%

PROJECTS

CoinVision -- Explainable AI Coin Classification Platform | *Python, PyTorch, ResNet18, FastAPI, SQLAlchemy, Grad-CAM*

- **Built an end-to-end image classification system using a fine-tuned ResNet18 model across 315+ coin categories and ~10,000 images**.
- Designed a multi-stage training pipeline with **data augmentation, stratified validation, checkpoint selection, and progressive fine-tuning**.
- Implemented **Grad-CAM explainability**, generating visual heatmaps alongside Top-3 predictions and confidence scores.
- Productionized model inference through a **FastAPI REST service** with SQLAlchemy-backed prediction history and an interactive web interface.

College Event Management Platform | *FastAPI, PostgreSQL, SQLAlchemy, Docker, JWT, GitHub Actions*

- **Engineered a role-based event management backend with FastAPI, PostgreSQL, and JWT authentication**, supporting event creation, participant registration, search, filtering, and pagination.
- Designed a layered **Routes-Services-Repositories architecture** with QR-based attendance, analytics dashboards, and automated email notifications.
- Containerized and deployed the application using **Docker**, PostgreSQL, and Alembic migrations; added automated testing with **Pytest** and CI/CD through **GitHub Actions**.

System Design & Architecture Analysis Platform | *Python, FastAPI, Pydantic, Graph Theory, REST APIs*

- **Designed a modular architecture-analysis platform using FastAPI and Pydantic**, separating API, schema, service, data-model, and analysis layers.
- Modeled software systems as **dependency graphs** to analyze component relationships, identify structural bottlenecks, and visualize architecture.
- Built REST APIs for architecture analysis and applied principles of **modularity, scalability, and maintainability** to support extensible system modeling.

Numerical Analysis Toolkit | *Python, NumPy, SciPy, PyQt5, Matplotlib*

- **Developed a modular scientific-computing toolkit implementing Gauss-Legendre quadrature, spectral collocation, polynomial analysis, and boundary-value problem solvers**.
- Built an interactive PyQt5 interface for **matrix computation, visualization, and numerical error analysis**.

SKILLS

Programming Languages: Python, C++, C, SQL, HTML, CSS

Backend & Databases: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, SQLite, Pydantic, JWT

Machine Learning: PyTorch, Computer Vision, Transfer Learning, ResNet, Explainable AI, NumPy, Pandas, SciPy, Matplotlib

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, System Design, Software Architecture

DevOps & Tools: Docker, Git, GitHub Actions, Linux, Pytest, Alembic, Jupyter

POSITIONS OF RESPONSIBILITY

Media & Public Relations Secretary | Head, Media Council, IIT Hyderabad

- **Led the Media Council and coordinated cross-functional teams across institute-wide media, branding, and communication initiatives**.
- Managed recruitment, delegation, and mentoring while establishing workflows and review processes to improve **team accountability and execution**.
- Coordinated with institute administration, student bodies, and external stakeholders to execute **campus-wide events and communication campaigns**.

Web Coordinator | Media Council, IIT Hyderabad

- **Developed and maintained responsive web pages for the IIT Hyderabad Gymkhana website**, collaborating with cross-functional teams on feature development and usability improvements.
- Implemented new functionality, resolved bugs, and optimized website performance, contributing to a **scalable and accessible web platform**.

RELEVANT COURSES

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Probability & Statistics, Linear Algebra, Numerical Methods, Discrete Mathematics, Theory of Computation

EXTRACURRICULAR

Member of NSO Cricket Team

Selected to represent my hostel in the MILAN Inter-Hostel Cricket Tournament at IIT Hyderabad.

Part of Gymkhana IITH 2026-27 team